CHAPTER 9

CHI-SQUARE TEST

CHAPTER ESSENCE AT A GLANCE:
Master Karl Pearson's Chi-Square distribution: Goodness of Fit test, Test of Independence of Attributes in contingency tables, Yates' correction for continuity, and conditions for validity.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Chi-Square Distribution Nature:** Asymmetric, positively skewed distribution defined for non-negative values [0, infinity); shape governed exclusively by degrees of freedom; Mean = df, Variance = 2 * df.

• **Goodness of Fit Test:** Compares observed frequencies (O) against theoretical expected frequencies (E); Chi-Square = sum [(O - E)^2 divided by E] with df = n - 1 - k (k = estimated parameters).

• **Independence of Attributes in Contingency Table:** Tests association between two categorical attributes; Expected cell frequency E = (Row Total * Column Total) divided by Grand Total; for r x c table, df = (r - 1)(c - 1).

• **Yates' Correction for Continuity:** Applied to 2x2 contingency tables when any cell frequency is small; subtracts 0.5 from absolute deviation: Chi-Square = sum [(|O - E| - 0.5)^2 divided by E] with df = 1.

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{(k-1)} \quad [\text{Goodness of Fit Statistic}]$$

Pearson Discrepancy Measure Between Observed and Expected

$$E_{ij} = \frac{R_i \times C_j}{N} \rightarrow \text{Expected Cell Frequency in Contingency Table}$$

Theoretical Expected Value Under Independence Hypothesis

$$\text{Contingency Table df} \Longleftrightarrow (r-1)(c-1) \quad [\text{for } r \times c \text{ Categories}]$$

Degrees of Freedom for Two-Way Categorical Association

$$\chi^2_{\text{Yates}} = \frac{N(|ad-bc| - \frac{N}{2})^2}{(a+b)(c+d)(a+c)(b+d)} \quad [2 \times 2 \text{ Table with df} = 1]$$

Yates Continuity Correction for Discrete Cell Frequencies

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Total expected frequency must ALWAYS equal total observed frequency: sum E = sum O = N.

• **Alert:** No expected cell frequency should be less than 5; if any expected frequency is less than 5, pool adjacent categories before computing Chi-Square.

• **Alert:** Total sample size N should be reasonably large (N >= 50) for Chi-Square approximation to be valid.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State any two essential conditions for the validity of the Chi-Square test. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Condition 1:** The total sample size N must be reasonably large, preferably at least 50. **[1 Mark]**

• **Condition 2:** No theoretical expected cell frequency should be less than 5; if less than 5, it must be pooled with an adjacent cell. **[1 Mark]**

2. A die is thrown 120 times and the frequencies of faces 1, 2, 3, 4, 5, 6 are observed to be: 15, 25, 18, 16, 22, 24. Test whether the die is fair at 5 percent level of significance (given critical Chi-Square for 5 df at 5 percent is 11.07). [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Hypotheses:** H0: The die is fair (faces are equally likely). H1: The die is biased. df = 6 - 1 = 5. Critical value = 11.07. **[1 Mark]**

• **Expected Frequencies:** E = 120 divided by 6 = 20 for each face. **[1 Mark]**

• **Chi-Square Calculation:** sum [(O - E)^2 divided by E] = [(15-20)^2 + (25-20)^2 + (18-20)^2 + (16-20)^2 + (22-20)^2 + (24-20)^2] divided by 20 = [25 + 25 + 4 + 16 + 4 + 16] divided by 20 = 90 divided by 20 = 4.50. **[1 Mark]**

• **Conclusion:** Since calculated Chi-Square = 4.50 < 11.07, we fail to reject H0. The die can be regarded as fair at 5 percent level. **[1 Mark]**

3. In an anti-malarial drive, 200 patients were examined. Out of 100 vaccinated patients, 20 were attacked by malaria; out of 100 unvaccinated patients, 40 were attacked. Test whether vaccination is effective in preventing malaria at 5 percent level (given critical Chi-Square for 1 df at 5 percent is 3.841). [Score: 4 Marks]

Answer & Key Valuation Steps:

• **2x2 Contingency Table:** Vaccinated Attacked a=20, Not Attacked b=80 (Total 100); Unvaccinated Attacked c=40, Not Attacked d=60 (Total 100). Column Totals: Attacked = 60, Not Attacked = 140. Grand Total N = 200. **[1 Mark]**

• **Expected Frequencies:** E(a) = (100 * 60) divided by 200 = 30; E(b) = (100 * 140) divided by 200 = 70; E(c) = 30; E(d) = 70. **[1 Mark]**

• **Chi-Square Calculation:** (20-30)^2 divided by 30 + (80-70)^2 divided by 70 + (40-30)^2 divided by 30 + (60-70)^2 divided by 70 = 100 divided by 30 + 100 divided by 70 + 100 divided by 30 + 100 divided by 70 = 3.33 + 1.43 + 3.33 + 1.43 = 9.52. **[1 Mark]**

• **Conclusion:** Since calculated Chi-Square = 9.52 > 3.841, reject H0. Vaccination and incidence of malaria are significantly associated; vaccination is effective. **[1 Mark]**

4. What is Yates' Correction for Continuity in a Chi-Square test? When and why is it applied? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **When Applied:** Applied to 2x2 contingency tables with 1 degree of freedom when cell frequencies are small (less than 5). **[1 Mark]**

• **Why Applied:** Chi-Square is a continuous theoretical distribution, whereas cell frequencies are discrete integers; Yates' correction adjusts for this continuity mismatch. **[1 Mark]**

• **How Calculated:** Subtracts 0.5 from the absolute difference between observed and expected frequencies before squaring: (|O - E| - 0.5)^2 divided by E. **[1 Mark]**

5. What is the number of degrees of freedom in a 3 x 4 contingency table? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Formula:** df = (r - 1) * (c - 1), where r = number of rows and c = number of columns. **[1 Mark]**

• **Calculation:** df = (3 - 1) * (4 - 1) = 2 * 3 = 6. **[1 Mark]**

6. State the Mean and Variance of a Chi-Square distribution with n degrees of freedom. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Mean:** Mean = n (equal to its degrees of freedom). **[1 Mark]**

• **Variance:** Variance = 2n (twice its degrees of freedom). **[1 Mark]**